

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) is a powerful framework for enhancing the reasoning capabilities of large language models. Yet existing approaches such as GRPO are fundamentally hampered by zero gradients, a failure that stems directly from two design flaws: fixed clipping bounds for token-level probability ratios and the standardization of identical rewards. Together these flaws produce ineffective gradient updates and waste a large fraction of generated responses. We resolve both problems with **Dynamic Clipping Policy Optimization (DCPO)**, which combines a dynamic clipping strategy that adaptively adjusts clipping bounds according to token-specific prior probabilities to unlock token-level exploration, with a smooth advantage standardization technique that standardizes rewards across cumulative training steps to fully exploit every generated response. DCPO sets a new state of the art across four benchmarks and four different models. On the Qwen2.5-Math-7B model it reaches an Avg@1 of 46.7 under greedy decoding and an Avg@32 of 38.8 under 32-times sampling on AIME24, decisively surpassing DAPO (36.7/31.6), GRPO (36.7/32.1) and GSPO (40.0/34.9). On AIME25 with Qwen2.5-14B, DCPO attains (23.3/19.0), far ahead of GRPO (13.3/10.5), DAPO (20.0/15.3) and GSPO (16.7/9.9). Beyond accuracy, DCPO delivers an average 28% gain in nonzero advantage over GRPO across all four models, doubles training efficiency relative to DAPO, and cuts the token clipping ratio by an order of magnitude compared to both GRPO and DAPO—all while achieving superior performance. These results establish DCPO as a markedly more efficient way to leverage generated data for reinforcement learning in large language models.

1 INTRODUCTION

The reasoning capabilities of large language models (LLMs) have become a central focus of recent research (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Reinforcement Learning from Verifiable Rewards (RLVR) is a leading technique for driving these capabilities forward. By optimizing LLMs with rule-based outcome rewards, RLVR delivers clear gains in domains that demand reflective decision-making, such as mathematics and coding (DeepSeek-AI et al., 2024).

The GRPO (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) family of algorithms was developed to support RLVR. While GRPO’s performance has been validated across many studies, recent work such as DAPO (Yu et al., 2025) exposes critical and recurring limitations. GRPO suffers from entropy collapse, with advantage values decaying to zero in later training stages. Compounding this, inappropriate weight-clipping discards rarely sampled learning signals and sharply curtails the diversity of model exploration.

DAPO attempts to patch these issues with the Clip-Higher strategy, which relaxes the upper clipping bound to counter entropy collapse, and the Dynamic Sampling strategy, which filters out samples with uniform rewards so that every batch contains only samples with valid gradients. These fixes, however, leave two decisive problems unsolved: (i) **Response-Level Inefficiency**: dynamic sampling inherently discards data and slows training convergence. (ii) **Token Clipping Deficit**: our analysis shows that the optimal clipping bound varies from token to token, so any uniform upper bound is inherently suboptimal. Methods that avoid directly discarding zero-gradient responses still fail to escape these limitations.

We introduce **Dynamic Clipping Policy Optimization (DCPO)**, which directly resolves the key limitations of previous methods. First, we introduce a **smoothing technique that standardizes**

advantages across cumulative steps to eliminate the gradient-vanishing problem caused by zeroing out the advantages of responses with the same reward. By aggregating the reward distribution of all cumulatively generated responses with the current distribution, this method unlocks far more efficient response-level exploration (see Section 2). Second, we propose a **dynamic clipping method that adaptively adjusts the clipping bounds based on the old token-specific probabilities**, replacing the crude device of fixed upper and lower clipping bounds. Concretely, tokens with lower prior probabilities are granted wider clipping bounds, opening up substantially more room for token-level exploration (see Section 2 and Appendix A.6). In addition, DCPO computes the mean loss only across tokens within each individual response rather than averaging over an entire batch, which preserves the relative advantage structure among responses to the same prompt and eliminates the dilution effect of batch-level averaging.

Finally, we merged Math-DAPO-17k (Yu et al., 2025) with levels 3-5 of the MATH dataset (Liu et al., 2025) as the training dataset and selected four common mathematical reasoning benchmarks for evaluation. We ran experiments on four different models against GRPO, DAPO and (where available) GSPO as baselines. DCPO consistently wins, with especially large gains on the challenging AIME benchmarks: on AIME24-Avg@32 with 32-time sampling, DCPO-7B scores 38.8 vs. 32.1 (GRPO) vs. 31.6 (DAPO) vs. 34.9 (GSPO), and on AIME25-Avg@32, DCPO-14B reaches 19.0 vs. 10.5 (GRPO) vs. 15.3 (DAPO) vs. 9.9 (GSPO). DCPO also raises the average nonzero advantage ratio by 28% over GRPO, doubles training efficiency over DAPO, and cuts token clipping by an order of magnitude compared to all baselines, all while delivering superior performance.

For preliminary details, including loss and advantage calculation methods for previous algorithms such as GRPO, DAPO and GSPO, see the Appendix A.2.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

Clipping the probability ratio is a standard technique in reinforcement learning for stabilizing policy updates and preventing large, destabilizing parameter shifts during optimization. It is especially critical when applying importance sampling in off-policy settings. Reweighting by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$ yields an unbiased estimate of $\mathbb{E}_{q(x)}[f(x) \frac{p(x)}{q(x)}]$, but at the cost of a sharply inflated variance, as Equation (1) makes explicit.

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (1)$$

To contain this variance inflation, prior works (e.g., Schulman et al. (2017)) constrain the probability ratio within a fixed clipping bound governed by the hyperparameter ϵ as Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

A symmetric fixed clipping bound, however, throttles the model’s exploration of the diversity space, most severely in regions where the old policy assigns low probability mass. There, the absolute bound for decreasing probabilities is inherently tiny, which strangles any capacity for meaningful policy updates. DAPO’s asymmetric clipping bounds do not fix this—the fundamental limitation persists. GSPO makes matters worse: it discards entire responses whenever the sequence-level importance ratio exceeds the asymmetric clipping bounds, wasting informative tokens while retaining high-variance ones and thereby destabilizing training.

The right approach, from the standpoint of controlling variance-bias, is to impose probability-dependent constraints, as formalized in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting $p(x) = r(x)q(x)$ into Equation (3) and enforcing non-negativity of the square-root term by taking its maximum with zero yields a closed-form solution for $r(x)$. This gives our **dynamic-adaptive clipping bounds**, which vary adaptively with $q(x)$ and the hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4).

This dynamic-adaptive clipping method unlocks substantially greater policy exploration in low-probability regions. Furthermore, inspired by the dual clipping method (Ye et al., 2020), we cap

$$0.5 + \frac{1}{2} \sqrt{\max \left(1 - \frac{4\epsilon_{low}}{q(x)}, 0 \right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

the maximum bounds for both positive and negative advantages at 10 to prevent excessive clipping bounds from overtraining. The full derivation and analysis of these dynamic clipping bounds from the importance sampling framework appear in Appendix A.5 and Appendix A.6.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

In prior works such as GRPO, DAPO and GSPO, the advantage $\hat{A}_{new,j}^i$ of the j -th response standardizes only the rewards $R_{j=1,\dots,G}^i$ for the same prompt at the i -th step, as in Equation (11) in Appendix A.3. This design has two damaging consequences. First, whenever randomness in response sampling makes all rewards identical at a given step, the advantage collapses to zero, so the prompt contributes nothing to parameter updates despite potentially valuable differences in reasoning trajectories. Second, randomness in high-entropy sampling produces highly skewed label counts, which drive large fluctuations in standardized advantage values across steps—even sign reversals—and destabilize training.

Recent work (Yue et al., 2025) further establishes that RLVR primarily refines a policy by adjusting the likelihood of correct samples rather than fundamentally altering the existing distribution. It follows that the overall reward distribution of responses to the same prompt is best treated as drawn from a single global distribution throughout training, which directly motivates our *cumulative* standardization in Equation (5).

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the statistics are computed over all responses generated so far for the same prompt.

To suppress the fluctuations of both step-specific standardization ($\hat{A}_{new,j}^i$) and cumulative standardization ($\hat{A}_{total,j}^i$), we introduce two smoothing functions, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$, each a weighted average of the two standardization methods whose weights shift with step i in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

$\hat{S}A_{new,j}^i$ increasingly weights the current distribution as training progresses, adapting to the latest policy outputs, whereas $\hat{S}A_{total,j}^i$ increasingly weights the cumulative distribution, treating the current step’s deviation as a fine-tuning adjustment within the broader distribution. We then define the final advantage $\hat{A}_j^i$ as the smoothed advantage with the smaller absolute value, which decisively curbs the effect of fluctuations from both cumulative and current-step standardization on training stability, as given in Equation (7).

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

Once a prompt enters optimization, the responses it generates in subsequent steps keep contributing to model optimization whether or not the current advantage is 0. As a result, even when identical rewards occur within a step, these responses still update the model with the advantage $\frac{1}{i} \hat{A}_{total,j}^i$, preserving useful learning signals and sharply improving data efficiency.

3.1 OTM LOSS

The **Token-Level Mean loss(TLM)** (the loss Equation (13) used in DAPO) balances all token contributions, but in doing so it corrupts the relative relationship between responses of different

lengths. Consider two responses to the same prompt: response A has an advantage of 1 and a length of 500, while response B has an advantage of 0.5 and a length of 1500. Under TLM, the loss for A is weighted by $\frac{500}{500+1500} = 0.25$ and for B by $\frac{1500}{500+1500} = 0.75$, so A contributes $1 \times 0.25 = 0.25$ and B contributes $0.5 \times 0.75 = 0.375$. The lower-advantage response B thus exerts the larger influence (0.375 vs. 0.25)—exactly the wrong outcome.

The Sequence-Level Mean loss (SLM) (the loss Equation (12) used in GRPO), which averages across batches, is no better: it dampens the advantage magnitude, effectively shrinking it by a factor of G after the advantage computation. With large batch sizes this destroys the correlations encoded in the original standardized advantages. Standardized advantages already carry the correct relative relationships across responses; averaging over the $|o_i|$ tokens within a single response preserves that structure while distributing importance equally across the tokens of that response. We therefore average only over the tokens within one response and never across responses in the same batch. We call this the Only Token Mean loss (OTM), given in Equation (8).

$$\mathcal{T}_{\text{DCPO}}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (8)$$

As in DAPO, the DCPO loss is free of any KL-divergence constraint. For the lower and upper clipping bounds, we use the dynamic clipping bounds introduced in Section 2.

4 EXPERIMENTAL SETUP

Models and Datasets We evaluate on four distinct models—Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base) and Qwen2.5-14B (common base) (Yang et al., 2024a;b)—using a merged training dataset of approximately 25k mathematical problems that combines the DAPO-Math-17K corpus (Yu et al., 2025) with levels 3-5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025). All problems are processed with the Qwen-Math template (see Section A.8) for both training and evaluation.

Evaluation We evaluate our models on four widely used mathematical reasoning benchmarks: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). For **MATH500** we report only Avg@1, given its larger size (500 problems). For the smaller **AMC23** (40 problems), **AIME24** (30 problems), and **AIME25** (30 problems) datasets, we report both Avg@1 and Avg@32 to guarantee a robust and comprehensive evaluation.

- **Avg@1**: standard accuracy under greedy decoding, measuring the model’s best single prediction.
- **Avg@32**: average accuracy over 32 sampled responses per problem at temperature 1.0 and top-p 1.0, directly probing the robustness and stability of the trained policy distribution.

Baseline Settings We use GRPO and DAPO as baselines and add GSPO on Qwen2.5-Math-7B and Qwen2.5-14B as additional, stronger baselines. The remaining settings—the various clipping thresholds ϵ_* , the codebase, and the common training hyperparameters—are detailed in Appendix A.9.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1 summarizes the performance of GRPO, DAPO, GSPO, and our proposed DCPO on four math reasoning benchmarks: MATH500, AMC23, AIME24, and AIME25. Reporting both greedy decoding (Avg@1) and sampling-based decoding (Avg@32) gives a comprehensive picture of each method’s robustness and generalization. Across every model scale, DCPO matches or surpasses GRPO, DAPO and (where available) GSPO, confirming its effectiveness as a general-purpose preference optimization technique for reinforcement learning of LLMs.

Overall Superiority of DCPO On Qwen2.5-Math-1.5B-Instruct, DCPO posts the highest average scores (47.2/32.8), beating GRPO (+1.2/+0.8) and DAPO (+0.7/+0.4). On Qwen2.5-3B, DCPO reaches (37.6/22.7), outscoring GRPO (+1.3/+1.7) and leading DAPO on Avg@1 (+1.0) with only a marginal Avg@32 gap (-0.4). On Qwen2.5-Math-7B, DCPO attains (57.1/45.2), sweeping past

Table 1: Performance across benchmarks. Avg@1: greedy decoding; Avg@32: sampling 32 times. Boldface marks the best value in each column, and Δ reports DCPO’s improvement over each baseline.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

GRPO (+4.0/+3.6), DAPO (+3.3/+2.8) and GSPO (+1.9/+1.9). On the largest tested model, Qwen2.5-14B, DCPO delivers a commanding (53.2/39.0), with large margins over GRPO (+6.5/+7.7) and clear wins over DAPO (+1.3/+0.1) and GSPO (+4.2/+5.5) on Avg@1 and Avg@32. These results establish that DCPO improves robustly across model capacities and aligns to the reward signal better than every baseline under both greedy and stochastic decoding.

Robustness on AIME under Sampling (Avg@32) The most striking trend appears on AIME24 and AIME25, high-difficulty competition math benchmarks built to stress-test reasoning ability. Under Avg@32, which measures a model’s ability to generate correct responses under high-entropy sampling, DCPO is markedly more robust at the 7B and 14B scales: on AIME24-Avg@32, DCPO-7B achieves 38.8, well ahead of GRPO (32.1), DAPO (31.6) and GSPO (34.9); on AIME25-Avg@32, DCPO-14B reaches 19.0, far exceeding GRPO (10.5), DAPO (15.3) and GSPO (9.9). These gains show that DCPO improves not only average-case performance but also the diversity and reliability of correct generations—precisely what matters under sampling-based decoding with beam search or temperature-driven variability. By optimizing reward fidelity and exploration together under high-entropy decoding, DCPO proves its practical value in real-world settings where multiple plausible outputs are sampled.

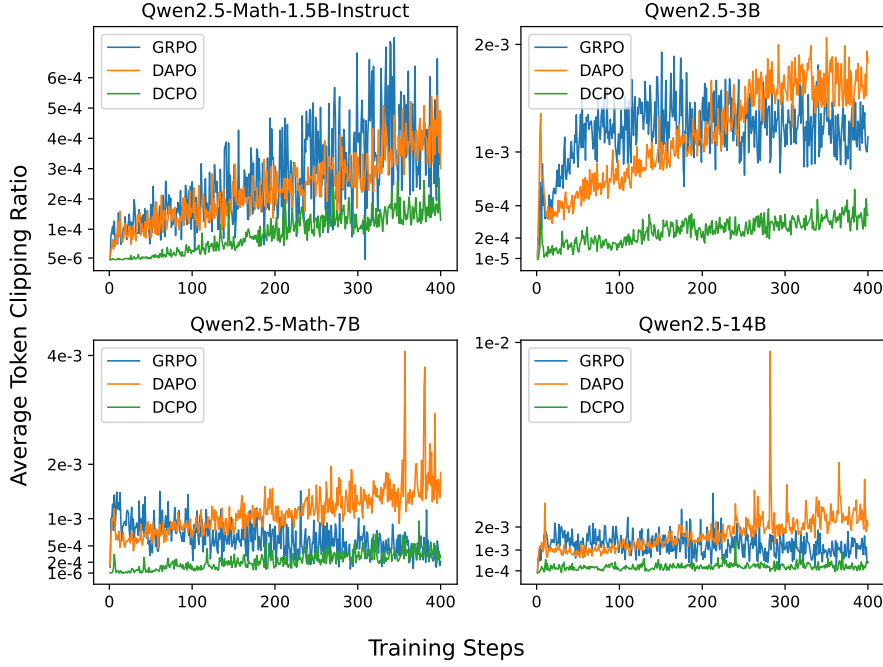


Figure 1: *TCR* across models and methods. DCPO holds the *TCR* an order of magnitude below GRPO and DAPO throughout training.

5.2 THE TOKEN CLIPPING RATIO (*TCR*)

We define the **Token Clipping Ratio** (*TCR*) as the proportion of tokens excluded from policy updates during back-propagation due to clipping. Figure 1 shows that *TCR* behavior differs dramatically across reinforcement learning methods.

$$\text{TCR} = \text{Average} \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } \textit{micro}_m}{\text{Total number of tokens in } \textit{micro}_m} \right) \quad (9)$$

where $\textit{micro}_m$ denotes the m -th microbatch and N is the number of microbatches per training step.

In GRPO training, *TCR* splits by model size: for smaller models (Qwen2.5-Math-1.5B and Qwen2.5-3B) it rises with training steps, while for larger models (Qwen2.5-Math-7B and Qwen2.5-14B) it gradually falls. DAPO, by contrast, drives *TCR* upward for every model, so as training proceeds it clips ever more tokens and increasingly bases policy updates on partial or truncated responses. Worse, both GRPO and DAPO show large fluctuations and occasional abnormal *TCR* spikes, which inject instability into optimization and can trigger entropy collapse. GSPO fares even worse on the larger models: its average *TCR* exceeds 11% on Qwen2.5-Math-7B and 15% on Qwen2.5-14B, far above the token-level clipping methods, and it discards far more informative tokens as a result.

DCPO stands apart, holding a remarkably stable and efficient training state. Whatever the model size or training stage, DCPO’s *TCR* stays nearly constant and an order of magnitude below GRPO and DAPO. This consistently lower *TCR* means DCPO discards far fewer tokens, lets a much larger share of complete response sequences drive policy updates, and frees up room for the model to explore the diversity of rare tokens. DCPO thus achieves clearly superior sample efficiency while keeping outlier variation tightly controlled.

Take Qwen2.5-Math-7B: the token distribution rapidly concentrates on high-confidence tokens—after roughly 60 steps about 95% of generated tokens satisfy $q(x) > 0.9$, rising to about 97% after roughly 100 steps and climbing further in later training. As Figure 4d shows, once the old probability satisfies $(q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83)$, the new probability $p(x)$ contributes to model updates within the interval $[\frac{1}{1+\epsilon}, 1]$ and is never discarded under any clipping method. Because the vast majority of

tokens fall in this never-clipped region, the low overall TCR we observe is fully explained. Meanwhile, Figure 1 shows TCR growing in the later phases of most training runs, which means the baselines discard a growing fraction of low-probability tokens and steadily choke off the model’s ability to explore rare tokens.

Wang et al. (2025) established that **high-entropy tokens**—precisely the low-probability tokens—are the primary drivers of a model’s emerging reasoning capabilities. Since entropy and probability are inversely related, the rarer a token, the more information it carries. Granting a broader admissible update range to such low-probability tokens therefore enlarges the model’s effective exploration space and accelerates the acquisition of diverse reasoning capabilities.

These empirical and theoretical findings directly justify our **dynamic-adaptive clipping** scheme, which relaxes clipping for tokens with $q(x) < \frac{1}{1+\epsilon}$ while preserving the safe interval for high-confidence tokens. By adaptively widening the update interval for the high-entropy tail of the distribution, the method restores the model’s capacity to learn from rare, informative tokens and thereby strengthens its reasoning ability.

5.3 THE RESPONSE UTILIZATION RATIO (RUR)

During RLVR training, a policy update requires a nonzero gradient for a given response. We quantify this efficiency with the **Response Utilization Ratio** (RUR), the percentage of generated responses with nonzero advantage that participate in the policy update.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

Table 2 shows that GRPO’s average RUR is strikingly low, only 44.6% across the four models. GRPO therefore throws away more than half of every batch of generated responses—a severe inefficiency in response utilization that cripples diversity exploration. DCPO, in contrast, sustains an average RUR of roughly 70% after the first epoch: an absolute gain of 28% and a relative gain of 64% over GRPO.

Table 2: Average RUR over 400 training steps. DCPO nearly doubles the response utilization of GRPO and GSPO on every model.

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 makes the contrast vivid: GRPO’s RUR decays sharply over time for every model size and type, plunging from above 90% to below 50%. On Qwen2.5-Math-7B it collapses to as low as 30% after roughly 200 steps. GRPO’s reliance on current-step advantage standardization thus manufactures a large fraction of zero gradients and renders its diversity exploration deeply inefficient in later training. DCPO’s smooth advantage standardization removes this failure mode: after the first epoch its RUR quickly stabilizes around 70% and keeps improving with further training.

The RUR exceeds 90% in the first few steps but settles around 70% from the second epoch onward, never reaching 100%. Both behaviors have clear explanations, and together they reveal the real value of the SAS mechanism.

First, the initial RUR above 90% comes largely from early-stage outputs that violate the required formatting, incurring negative rewards and inflating RUR . This confirms that models quickly master fixed answer patterns, so later training should concentrate on their reasoning capabilities. Second, two factors keep the advantage consistently zero and hold RUR near 70% rather than 100%: (i) nearly half of these problems are easy enough that the model solves them reliably, and excluding them from updates is entirely appropriate because the model’s core capabilities need no further refinement there; (ii) the remaining cases are consistently wrong, for two reasons. Some problems are simply too hard for the current model even with a correct ground-truth label. Others reflect the DAPO-Math-17K dataset itself, which contains items whose labels were converted to mislabeled integers, introducing spurious “incorrect” labels. In both cases, discarding zero-advantage responses acts as a beneficial filter that shields the model from noisy or mislabeled data.

Unlike GRPO and DCPO, DAPO discards responses with identical labels from generated data, which precludes a direct *RUR* comparison. We therefore gauge DAPO’s training efficiency by the number of generated responses and the training time it needs to complete the same number of update steps. To offset the low sampling efficiency that GRPO’s low *RUR* would otherwise impose, we set DAPO’s generation batch size at 1-4 times the training batch size and adjust it dynamically. Across all four models, DAPO needs 3 to 5 times more generated responses than DCPO to reach the same number of parameter updates, which at least doubles the training GPU hours—and still yields no better performance. DAPO’s data utilization also deteriorates as models grow: on Qwen2.5-14B, the fraction of responses it uses falls below 30% after about 300 steps. DAPO’s dynamic sampling is thus decisively inefficient in both data utilization and training time, and it forfeits valuable exploration of model diversity.

In short, DCPO is a decisive advance over both GRPO and DAPO: it substantially raises response utilization and training efficiency, exploits every sampled response to open up more exploration space for model diversity, and lifts final performance.

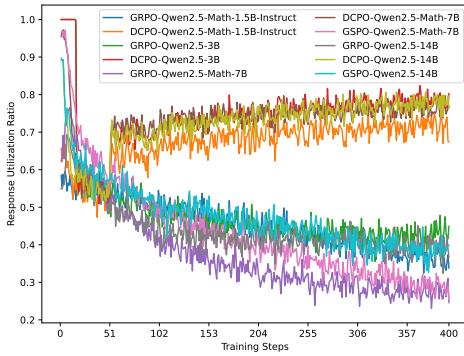


Figure 2: *RUR* progression during training. DCPO holds a high, stable *RUR* while GRPO’s collapses.

5.4 ABLATION STUDIES

We ablate DCPO on Qwen2.5-Math-7B to isolate the contribution of each component, using **Avg@32** as the metric because it directly exposes the robustness and stability of the learned policy distribution. For a fair comparison, each experiment modifies a single component of the baseline GRPO framework while holding all other settings fixed and removing the KL divergence term to align with DAPO, GSPO, and the full DCPO. The learning curves from 20 to 400 steps appear in Figure 3.

The curves for all three individual components show the model steadily improving its inference ability, surpassing GRPO and matching or exceeding DAPO and GSPO. Every modification delivers a distinct benefit, and together they produce a large overall gain over the baselines, demonstrating the effectiveness of the DCPO pipeline for reinforcement learning. In later training the baselines plateau and stop improving, whereas DCPO keeps trending upward, both per component and overall. The ablation confirms that each component improves training efficiency and that their integration compounds these gains into greater accuracy and stability. The details follow.

(i) **Only-Token-Mean loss (OTM)**: aggregating the loss within a single response rather than across the whole batch produces a model that clearly outperforms the baselines, which suffer from lower data utilization and fluctuating performance. (ii) **Smoothed Advantage Standardization (SAS)**: replacing the stepwise advantage with the cumulative smoothed advantage markedly improves GRPO and closes in on DAPO, proving that aggregating reward statistics across training steps stabilizes optimization and raises final performance. (iii) **Dynamic Adaptive Clipping (DAC)**: replacing GRPO’s fixed clipping boundary with a probabilistically adaptive one yields large improvements; DAC alone surpasses GRPO, DAPO, and GSPO and is more stable than OTM alone. (iv) **Combination of OTM + SAS + DAC (full DCPO)**: combining all three mechanisms achieves the best and most stable results overall, decisively outperforming GRPO, DAPO, and GSPO. The synergy among OTM, SAS, and DAC shows that each design choice contributes uniquely to data-efficient, stable reinforcement learning and to stronger, more robust large language models.

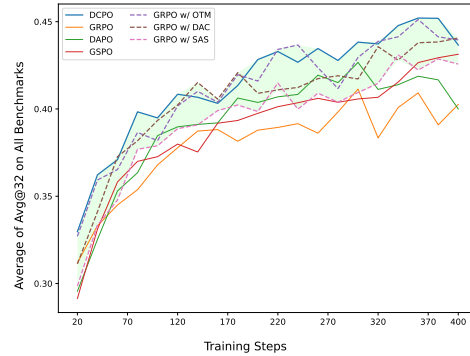


Figure 3: Ablation using the average Avg@32 on Qwen2.5-Math-7B. Each DCPO component adds a distinct, additive gain.

Overall, the ablation confirms that every DCPO component contributes positively and that their combination compounds into substantial cumulative gains, validating the proposed mechanisms for improving data efficiency and stability in reinforcement learning for LLMs.

6 CONCLUSION

This work resolves two key limitations of existing policy optimization methods: the restricted token-level exploration caused by fixed clipping in low-probability regions, and the low sample efficiency caused by current-step reward standardization. We introduced **Dynamic Clipping Policy Optimization (DCPO)**, a reinforcement learning pipeline that strengthens the reasoning capabilities of large language models through two innovations: (i) a **dynamically adaptive token-level clipping** mechanism that adjusts bounds according to old-policy probabilities, unlocking efficient exploration of rare, low-probability tokens, and (ii) a **Cumulative-Smooth Advantage Standardization** that aggregates the cumulative reward distribution across optimization steps, neutralizing the reward randomness of high-entropy sampling. Extensive experiments across four mathematical reasoning benchmarks and four models establish DCPO’s effectiveness: the DCPO-7B variant reaches an Avg@32 of 38.8 on the challenging AIME24 benchmark, clearly outperforming GRPO (32.1) and DAPO (31.6). DCPO further raises response utilization by 28% over GRPO, doubles training efficiency relative to DAPO, and cuts the token clipping ratio by an order of magnitude. Together these results demonstrate that DCPO substantially improves data utilization and diversity exploration in RLVR. Future work will extend DCPO to other domains, such as code generation and semantic reasoning.

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A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_{\theta}(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_{\theta}(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_{\theta} \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

In previous works, including GRPO and DAPO, the advantage $\hat{A}_{j,t}^i$ for token t in response j is computed by standardizing the reward R_j^i against the mean μ^i and standard deviation σ^i of the rewards of the G responses generated at the i -th step, as in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

When the rewards for the same prompt are identical, responses with zero advantage contribute nothing to the model update, wasting those responses outright.

A.3.1 GRPO

Shao et al. (2024) proposed the **Group Relative Policy Optimization (GRPO)**, which first samples G responses for each query, assigns rewards R through a rule-based reward function and estimates token-level advantages as in Equation (11). Finally, the model parameters are updated using Equation (12), referred to as **Sequence Level Mean loss (SLM)** as:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \quad (12)$$

The first term averages token-level advantages over all tokens in a response and then over all G responses in the batch. The $\mathbb{D}_{\text{KL}}$ penalty keeps the policy from drifting too far from a reference policy (e.g., a pre-trained model), and the fixed clipping threshold ε blocks overly large updates by capping the change in token probability ratios. Widely adopted as it is, this averaging strategy inherently dilutes per-response importance.

A.3.2 DAPO

Yu et al. (2025) propose **Dynamic sAmpling Policy Optimization (DAPO)** after observing that in GRPO longer sequences exert a disproportionate influence on the gradient update. DAPO standardizes token counts across the entire batch, yielding the **Token Level Mean loss (TLM)** in Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. DAPO discards every prompt whose responses share an identical reward and regenerates responses to preserve batch size, which incurs severe data inefficiency.

A.3.3 GSPO

Zheng et al. (2025) propose **Group Sequence Policy Optimization (GSPO)**, arguing that token-level clipping injects high-variance noise into gradient estimates—noise that accumulates over long sequences and is further amplified by the clipping operation itself. GSPO addresses this by replacing token-wise clipping with sequence-wise clipping to obtain a more stable training signal.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

where $s_i(\theta) = \exp\left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log\left(\frac{\pi_{\theta}(o_{j,t}|q)}{\pi_{\theta_{old}}(o_{j,t}|q)}\right)\right)$

GSPO, however, discards more than 10% of responses that carry non-zero advantages, wasting substantial data and training budget. And while GRPO removes responses with sequence-level high variance, it still retains many responses with token-level high entropy, which destabilizes training.

A.4 REWARD CALCULATION

To provide a verifiable and interpretable supervision signal, we adopt a rule-based reward model that scores each generated response on answer correctness and format compliance, as in Equation (15). We assign +1 when both format and answer are correct, 0 when the format is correct but the answer is wrong, and -1 when the format is incorrect.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j is the index of the response group for a given problem in the i -th training step, and the reward is calculated based on the format and correctness of the response.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

Clipping the probability ratio is a crucial technique in reinforcement learning for stabilizing policy optimization and preventing large, destabilizing parameter updates. It is especially important when importance sampling is used to estimate expected values under a new policy from samples generated by an old policy.

The expected value of a function $f(x)$ under the new probability $p(x)$ can be rewritten as an expectation under the old probability $q(x)$ via the importance sampling weight, as shown in equation 16:

$$\int f(x)p(x) dx = \int f(x)\frac{p(x)}{q(x)}q(x) dx = \mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \quad (16)$$

This estimator is unbiased, but its variance can be sharply inflated—the well-known challenge of importance sampling. The true variance of $f(x)$ under the new policy is given by Equation (17), while the variance of the importance-sampled estimator under the old policy is given by Equation (18):

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p}[f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x)\frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

The difference between the importance-sampled variance and the true variance highlights the potential for inflation, as shown in Equation (19):

$$\text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

To mitigate this difference, many prior works, including PPO Schulman et al. (2017) and RLVR methods (DeepSeek-AI et al., 2024; Liu et al., 2025), employ a fixed symmetric clipping bound on the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

A fixed clipping bound, however, inherently constrains the policy’s capacity for exploration, and this is most damaging in regions where the old policy assigns low probability mass. There the absolute bound for low probabilities is severely limited, which restricts any meaningful policy update. DAPO’s asymmetric clipping bounds do not remove the fundamental limitation of a non-adaptive constraint.

Moreover, recent studies such as Yue et al. (2025) show that the reasoning paths RLVR models exploit are largely already present in the base model’s sampling distribution—further evidenced by the minimal parameter shift during off-policy batch updates and the correspondingly small fraction of clipped parameters. This stability motivates a more principled alternative: we constrain the probability ratio $r(x)$ through a dynamic-adaptive mechanism that folds the probability into the restriction, as in Equation (21), replacing the conventional fixed clipping of Equation (20).

$$\left| \left(\frac{p(x)}{q(x)} - 1 \right) p(x) \right| \leq \epsilon \quad (21)$$

Expanding the substitution of $p(x) = r(x)q(x)$ into the inequality of a previous equation yields the expression in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For this clipping ratio to be mathematically valid, it must respect the fundamental constraints of probability distributions and their ratios. As Equation (23) states, the probabilities $p(x)$ and $q(x)$ lie in $[0, 1]$, so the probability ratio $r(x)$ is non-negative and the clipping hyper-parameter ϵ is non-negative as well.

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

To ensure the validity and practical applicability of the clipping bound derived from Equation (22), we define the necessary conditions for its bounds as outlined in Equation (24).

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

As we can get from Equation (22) to Equation (24), the probability ratio $r(x)$ should be met by the following conditions:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

For the clipping bound, we consider the overlapping region centered around the ratio value of 1 to minimize the discrepancy of bounds between the two conditions and to maintain consistency with the conventional fixed bound of GRPO, and define the following conditions to ensure the validity of the clipping bound:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, to ensure that the square number is not less than zero, we use the method of comparing with 0 and taking the maximum value of the approximation like $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. Finally, we can get the following conditions, as **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2}\sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

For the upper and lower clipping bounds in Inequality 27, we set distinct upper and lower clipping thresholds ϵ_{high} and ϵ_{low} to align the clipping bound with GRPO at specific points.

A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

To make our dynamic-adaptive clipping bounds coincide with the fixed symmetric clipping bounds at the boundary points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as shown in Figure 4d, we derive the exact values of ϵ_{low} and ϵ_{high} by solving the equations in Equation (28).

$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \end{cases} \quad (28)$$

$$\begin{cases} q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2} \sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases}$$

Following common practice (e.g., GRPO) with a fixed clipping threshold $\epsilon = 0.2$, we obtain clipping hyperparameters of approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$. Here $\epsilon_{low} \neq \epsilon_{high}$ serves a fundamentally different purpose from the asymmetric fixed clipping of DAPO.

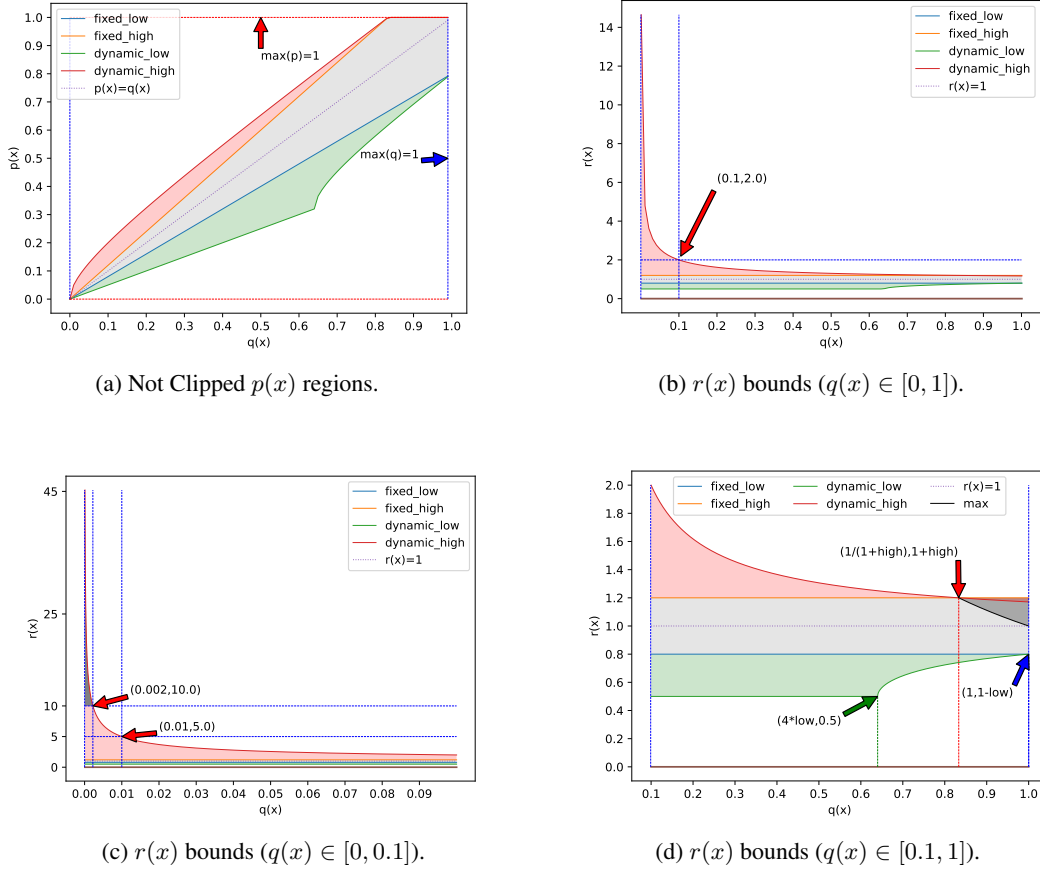


Figure 4: Clipping bound comparisons. Lines contrast fixed clipping ($\epsilon = 0.2$) with our dynamic-adaptive clipping ($\epsilon_{low} = 0.16, \epsilon_{high} = 0.2$), which widens the admissible region for rare tokens while matching the fixed bound for common ones.

Dynamic versus fixed clipping: Figure 4 contrasts the fixed clipping scheme of GRPO ($[1 - \epsilon, 1 + \epsilon]$) with our probability-aware adaptive clipping. For a token with old probability $q(x)$, the fixed bound gives a constant relative clipping width 2ϵ , so as $q(x)$ falls the absolute exploration space it permits shrinks linearly (Figure 4a). Our dynamic adaptation instead scales the relative width inversely with $q(x)$, keeping the exploration space roughly constant across the entire probability range.

Effect of decreasing $q(x)$: When $q(x)$ becomes small, the fixed scheme’s admissible interval $[1 - \epsilon, 1 + \epsilon]$ is independent of $q(x)$, so it does nothing to improve exploration of low-probability tokens. Our dynamic-adaptive bound instead expands rapidly (Figure 4b), granting a larger relative space to rare tokens—exactly the behavior efficient RL demands.

Behavior for extremely low probabilities: Figure 4c shows the clipping interval for $q(x) \in [0, 0.1]$. As $q(x) \rightarrow 0$, the dynamic-adaptive upper bound for $r(x)$ grows rapidly, inversely with the square

root, whereas the fixed-clipping value stays constant. Inspired by the dual-clipping strategy (Ye et al., 2020), we impose a hard ceiling $r_{\max} = 10$ to preclude the resulting instabilities (gradient vanishing, explosion, etc.). The gray region in Figure 4c corresponds to roughly $q(x) \leq 2 \times 10^{-3}$, where the adaptive upper limit is set to $r_{\max}$.

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\epsilon$ dynamic-adaptive clipping yields $dynamic_{low} = 0.5$, while fixed clipping holds the constant $fixed_{low} = 1 - \epsilon = 0.8$; as $q(x)$ rises past this threshold, the adaptive lower bound smoothly contracts toward the fixed value. Likewise, the adaptive upper bound exceeds the fixed bound whenever $q(x) < \frac{1}{1+\epsilon_{high}}$ and merges with it for larger $q(x)$. The adaptive interval $[dynamic_{low}(q(x), \epsilon_{low}), dynamic_{high}(q(x), \epsilon_{high})]$ therefore widens the admissible region for rare tokens in a probability-dependent way while preserving the tight fixed interval for common tokens (Figure 4d).

Positive versus negative advantage: The loss in Equation (8) contains both a positive and a negative advantage term. Under the min operation, the ratio $r(x)$ for a positive advantage is clipped to $[0, dynamic_{high}(q(x), \epsilon_{high})]$, and for a negative advantage to $[dynamic_{low}(q(x), \epsilon_{low}), \infty)$. Because an excessively large $r(x)$ would destabilize training, we cap the ratio for both positive and negative advantages at $r_{\max} = 10$. This precaution, inspired by Ye et al. (2020), guarantees that the adaptive limits never produce excessively high importance weights.

In short, for both positive and negative advantages, dynamic-adaptive clipping gives the model decisively more room to explore diversity precisely where rare tokens carry low probability—room that fixed clipping denies.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i denote the i -th iteration in which prompts generate responses and update the model parameters. We define the following statistics for reward normalization.

For the mean of the response rewards, we define the group of functions in Equation (29):

$$\begin{cases} \mu_{new}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{old}^i = \frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{total}^i = \frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{new}^i + (i-1) \mu_{old}^i) \end{cases} \quad (29)$$

For the variance of the response rewards, we define the group of functions in Equation (30):

$$\begin{cases} \sigma_{new}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{new}^i)^2} \\ \sigma_{old}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G (R_j^k - \mu_{old}^i)^2} \\ \sigma_{total}^i = \sqrt{\frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{total}^i)^2} \\ \quad = \sqrt{\frac{1}{i} \left(\sigma_{new}^i{}^2 + (i-1) \sigma_{old}^i{}^2 + \frac{i-1}{i} (\mu_{old}^i - \mu_{new}^i)^2 \right)} \end{cases} \quad (30)$$

To cut resource usage when computing σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29), we reuse the equivalent quantities $\mu_{new}^i, \sigma_{new}^i, \mu_{old}^i$ and σ_{old}^i rather than recomputing directly from the original rewards.

A.8 THE TEMPLATE FOR EXPERIMENTS

We use the following template, based on the Qwen-Math template (Yang et al., 2024b), for every experiment in this paper.

Qwen-Math Template

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant
```

A.9 TRAINING SETTING

We set a symmetric clipping threshold of $\epsilon = 0.2$ in GRPO and lower/upper clipping thresholds (0.2, 0.28) with an additional 512-token soft-punish cache and the maximum probability ratio 10 for negative advantage for DAPO, lower/upper clipping thresholds ($3e - 4, 4e - 4$) for GSPO, and low/high clipping thresholds (0.16, 0.2) for DCPO, preserving the same clipping points ($q(x), r(x) = (\frac{1}{1+\epsilon}, 1+\epsilon)$ and ($q(x), r(x) = (1, 1-\epsilon)$) as GRPO (Figure 4d), and set the maximum probability ratio to 10 for both positive and negative advantages, different from DAPO.

We train for 400 steps with a batch size of 512 for response generation and a mini-batch size of 32 for parameter updates. Using temperature 1 and top_p 1, we generate $G=16$ responses per problem, with the prompt max length set to 1,024 and the response max length to 3,072—far below the 20k maximum used in the original DAPO.

We adapt our training codebase from Verl (Sheng et al., 2024) on 32xH20 GPUs. Unless stated otherwise, hyperparameters match those of DAPO. For loss computation, we deliberately retain the original $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted-average formulation from GRPO or DAPO rather than the approximate weighting used in Verl, ensuring correctness across different microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

Avg@1 Trend Figure 5 reports the average Avg@1 metric, obtained by averaging the greedy-decoding (Avg@1) results on MATH500, AMC23, AIME24, and AIME25 to capture the model’s best performance. DCPO consistently outperforms GRPO and DAPO on all four models and beats GSPO on the two models where GSPO was evaluated. It is also markedly more stable than the baselines, free of the sharp drop seen in GRPO and the large fluctuations seen in DAPO. DCPO therefore elicits stronger reasoning capabilities during training and leaves more headroom for the model to reach even higher performance.

Avg@32 Trend Figure 6 plots the Avg@32 performance of DCPO against GRPO, DAPO and (where available) GSPO across model sizes on AMC23, AIME24, and AIME25, and it lays bare DCPO’s exceptional robustness and stability. GRPO, by contrast, fluctuates heavily and occasionally collapses—the Qwen2.5-14B model even suffers a sharp drop after roughly 170 steps. Across every evaluated model (1.5B to 14B), DCPO’s trajectory climbs steadily with minimal fluctuation, sustains the fastest early-stage growth, and keeps rising. And relative to DAPO, DCPO achieves superior performance while cutting training time and resources in half. These results confirm that dynamic-adaptive clipping and smooth advantage standardization jointly improve exploration at both the response level and the label level of diversity.

Entropy Trend Figure 7 shows GRPO’s entropy decaying sharply after an initial warm-up and quickly settling into a flat, low-entropy range. This entropy collapse is a rapid loss of policy randomness that shrinks the diversity of sampled responses and chokes off subsequent policy optimization.

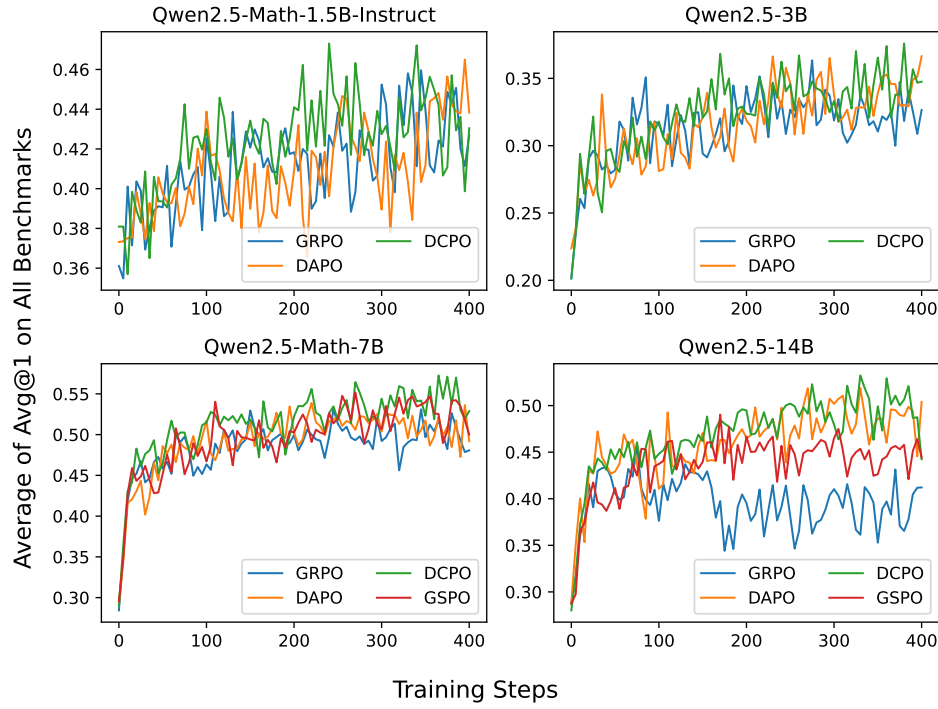


Figure 5: Avg@1 performance across benchmarks. DCPO leads every baseline on all four models with a smoother trajectory.

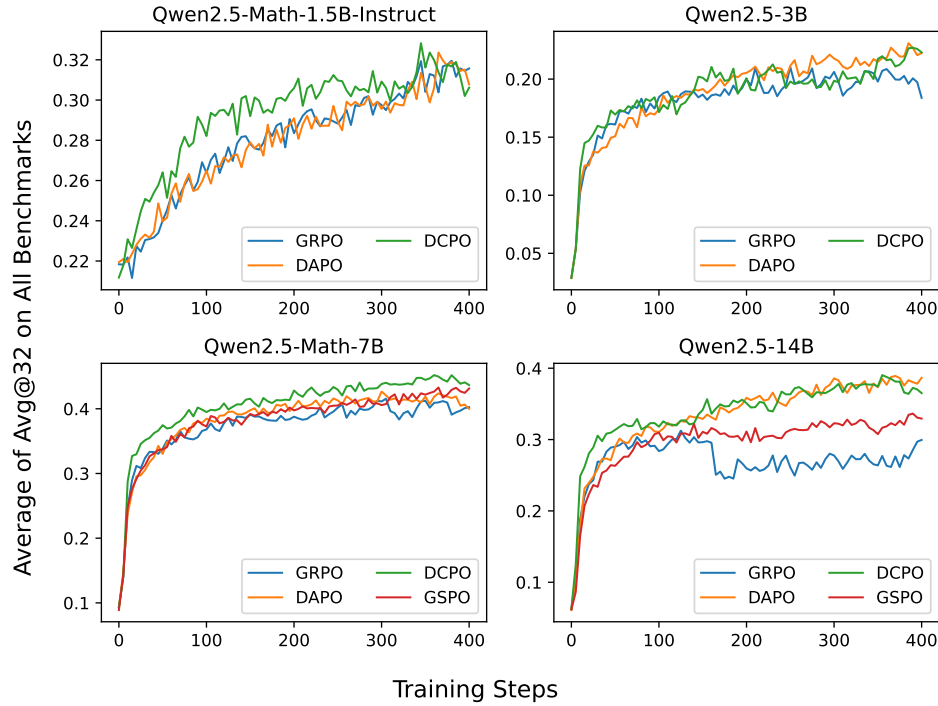


Figure 6: Avg@32 performance across benchmarks. DCPO trends steadily upward while GRPO fluctuates and collapses.

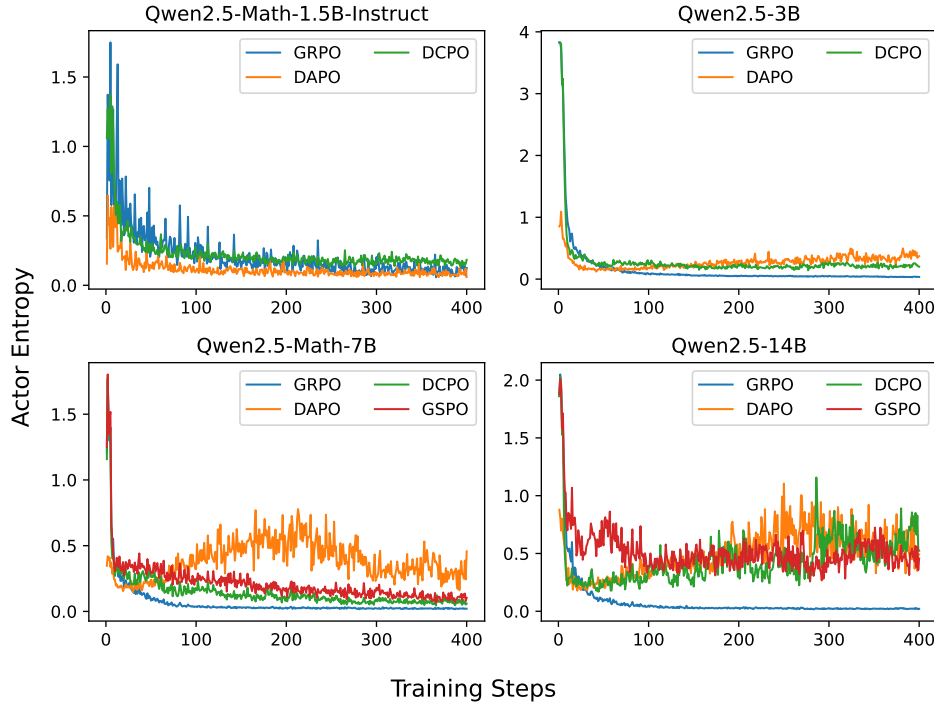


Figure 7: The entropy of the models during training. DCPO holds entropy in a moderate, well-balanced band.

DAPO and DCPO behave in a model-dependent way. On the Instruct model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO follow similar declining trajectories. On the larger base models (Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B), the trends diverge sharply: DAPO holds the highest entropy with a clear upward trend but large fluctuations. On Qwen2.5-Math-7B, GSPO’s entropy and average performance track DCPO closely; on Qwen2.5-14B, however, GSPO drifts downward with heavy fluctuation and performs poorly. DCPO instead settles at a moderate entropy level, with a markedly smoother curve confined to a bounded band between GRPO’s low-entropy region and DAPO’s high-variance region.

These observations reveal a U-shaped relationship between entropy and training performance: too little entropy starves the learner of exploratory signal and forces premature convergence, while too much or too erratic entropy wrecks training stability. DCPO holds entropy in a moderate, well-balanced range that secures both convergence and sufficient exploration, and this advantage grows with model capacity. DCPO’s entropy regulation is thus a central driver of its superior inference performance and enables more robust policy learning across model sizes.